THBKG: A Temporal Evaluation Harness for Target Identification

Computational Biology Symposium

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Date the evidence, not just the label: otherwise hindsight scores as foresight.

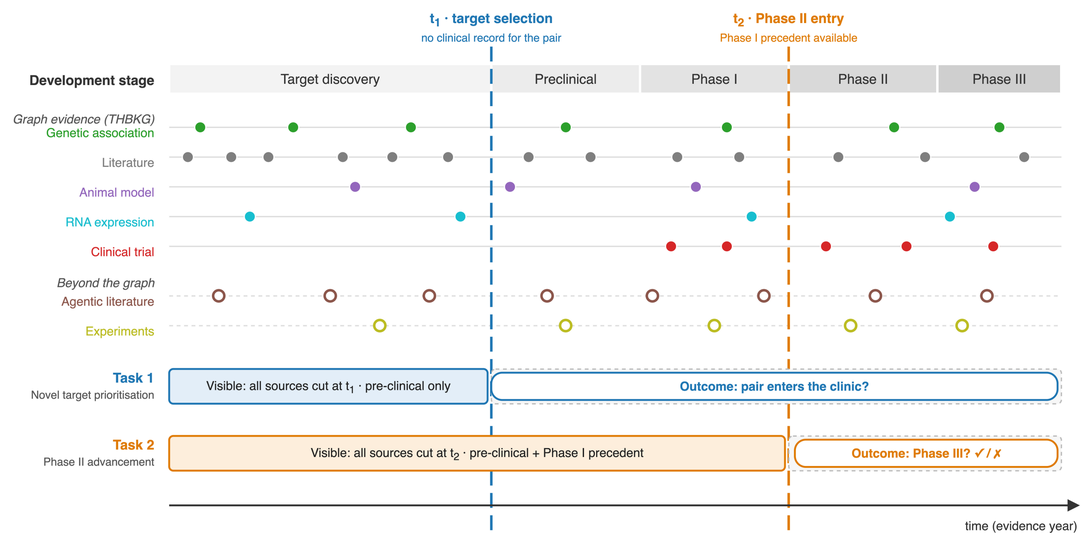
THBKG rewinds biomedical evidence to any decision point, tracing how far the intervening biology carries a hypothesis toward its clinical outcome.

Why Time Matters

Inadequate target–disease linkage accounts for 40–50% of Phase II efficacy failures, so anticipating which programmes will advance would let sponsors back the hypotheses most likely to reach patients. What a programme can be judged on is the evidence that supported its linkage *when it entered the clinic*.

- Target discovery platforms, **LLMs included, are benchmarked on how many already-known targets they recover**
- **Temporal leakage**, through the evidence and through the model's training cutoff, **inflates benchmark performance**
- The answer is decisive when model scores **steer which hypotheses enter costly trial and wet-lab validation**

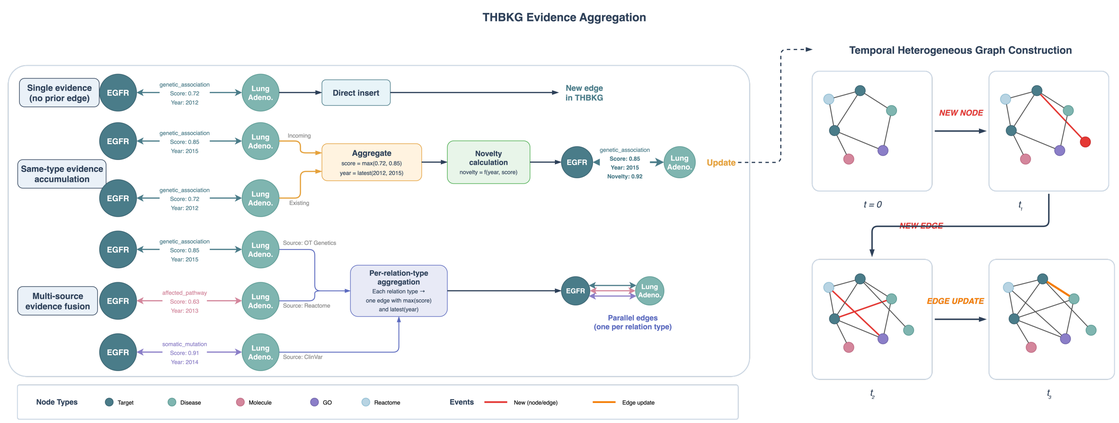
Evidence Through Time → Trial Decisions

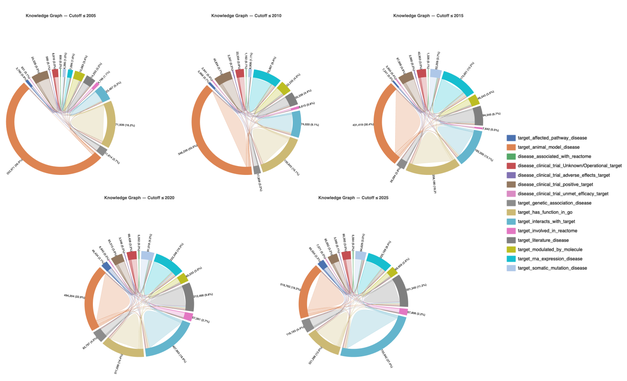


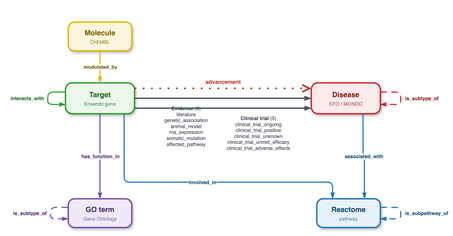
Every edge carries the year its evidence changed, so a pair's profile can be recovered as it stood *when its own decision fell due*. Two decisions, two tasks: at target selection (t_1), rank the targets each disease will newly enter trials for, using only evidence dated before that point; at Phase II entry (t_2), predict advancement to Phase III.

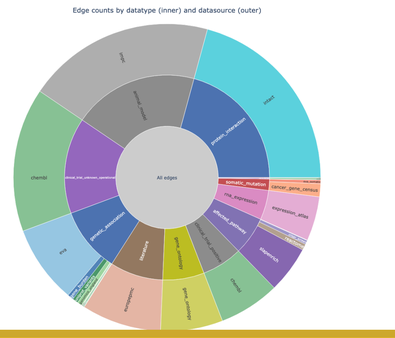
The THBKG Substrate

- **110,396 entities and 11.1M edges across nineteen relation types**, each edge carrying the year its evidence changed
- 5 node types (target, disease, molecule, Reactome pathway, GO term); **16 temporal + 3 static** ontology edge types; built from Open Targets, IntAct, GO and Reactome
- **As-of query**: $\mathcal{G}[T]$ returns the graph as it stood before date T, so hypotheses are judged on the evidence available at their decision date



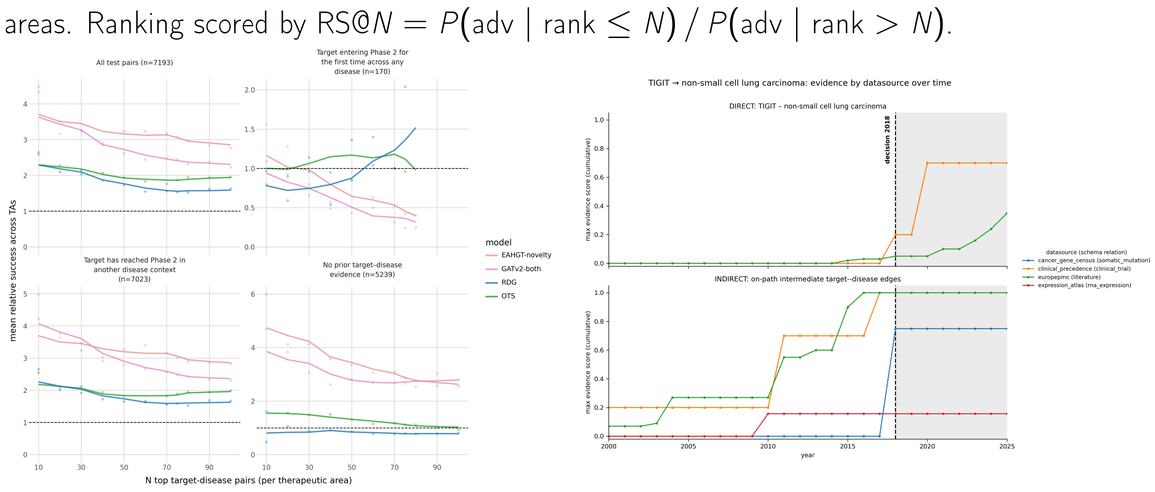


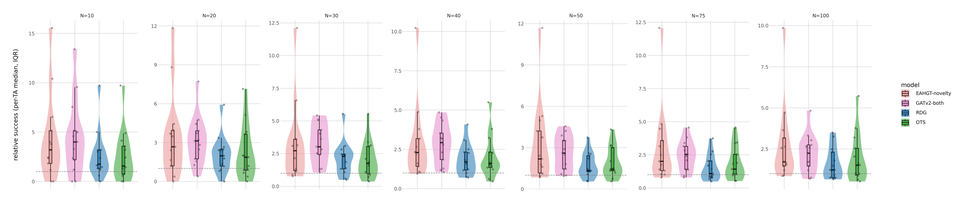




Will This Pair Advance to Phase III?

A decision-aligned benchmark: for a target–disease pair entering Phase II, predict whether it advances to Phase III on evidence datable before that decision; 7,193 pairs, 13 therapeutic areas. Ranking scored by RS@$N = P(\text{adv} \mid \text{rank} \le N) \,/\, P(\text{adv} \mid \text{rank} > N)$.

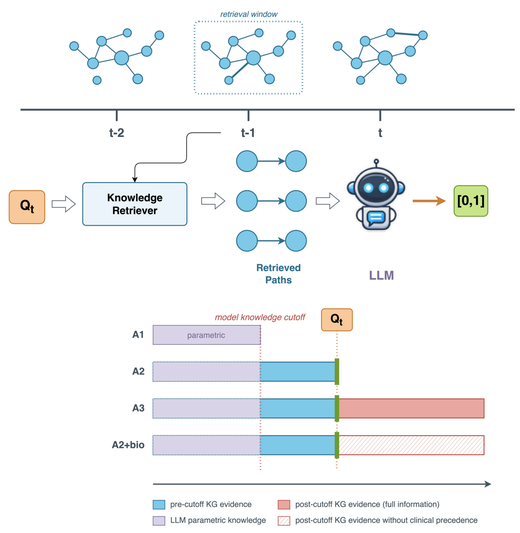


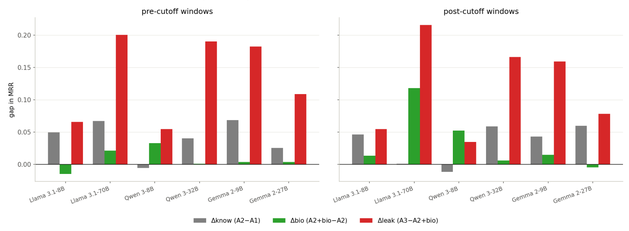


- **Outranks every direct-evidence reference**: RS@10 4.3–4.5 vs. 2.6 (TA mean)
- Gain concentrates on the **72.8% of pairs with no direct evidence, where a direct-edge model has nothing to read**: encoders rank five- to sixfold above chance via the intervening biology (Wilcoxon $p \le 0.004$)
- A **path-based explainer** decomposes each prediction into the evidence landscape behind the hypothesis

Did the LLM Forecast It, or Already Know It?

Six open LMs (Llama 3.1, Qwen 3, Gemma 2; 8B–70B) rank the targets a disease newly enters trials for (2021–2026), with THBKG rewound to each decision point under four retrieval conditions separating knowledge, fresh biology and leakage (left).





Model	MRR A1	MRR A2	MRR A2+bio	MRR A3	AUROC A1	AUROC A2	AUROC A2+bio	AUROC A3
Llama 3.1-8B	0.567	0.613	0.626	0.681	0.705	0.739	0.712	0.778
Llama 3.1-70B	0.620	0.621	0.739	0.955	0.741	0.760	0.798	0.971
Qwen 3-8B	0.592	0.581	0.633	0.668	0.730	0.705	0.731	0.762
Qwen 3-32B	0.627	0.686	0.692	0.858	0.763	0.801	0.801	0.928
Gemma 2-9B	0.606	0.649	0.664	0.824	0.737	0.745	0.749	0.894
Gemma 2-27B	0.620	0.680	0.675	0.754	0.764	0.758	0.767	0.855

Post-cutoff; chance MRR 0.38, AUROC 0.5. A1: the floor, what the model already knows; A2: honest decision-time evidence; A2+bio: fresh biology, no label; A3: full information (leaky).

- **That floor, not zero, is what retrieval has to beat**; honest evidence adds only a small lift, and it decays with recency
- Delete only the newest graph's post-cutoff trial edges and its advantage collapses
- **Verdict: recall of precedence, not foresight**

Takeaways

1. **First temporally indexed biomedical KG**: rebuilds the evidence landscape at any historical decision point
2. **Reasoning over the intervening biology**: propagation turns systems-level structure (pathways, GO functions, molecules) into pair-level advancement predictions
3. **Make the benchmark temporal**: each prediction evaluated against the evidence available at its decision time

References: Siu, Cabrera, Mudaliar, Zubiaga (2026) THBKG, under review; arXiv:2608.05982; Siu, Liang, Cabrera, Mudaliar, Zubiaga (2026) Did the Model Forecast It, or Already Know It? Under review, AI4DD Workshop, NeurIPS 2026; Czech et al. (2024) Clinical advancement forecasting, *medRxiv*; Ochoa et al. (2023) Open Targets Platform, *NAR*; Hu et al. (2020) Heterogeneous Graph Transformer, *WWW*.

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arXiv:2608.05982

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